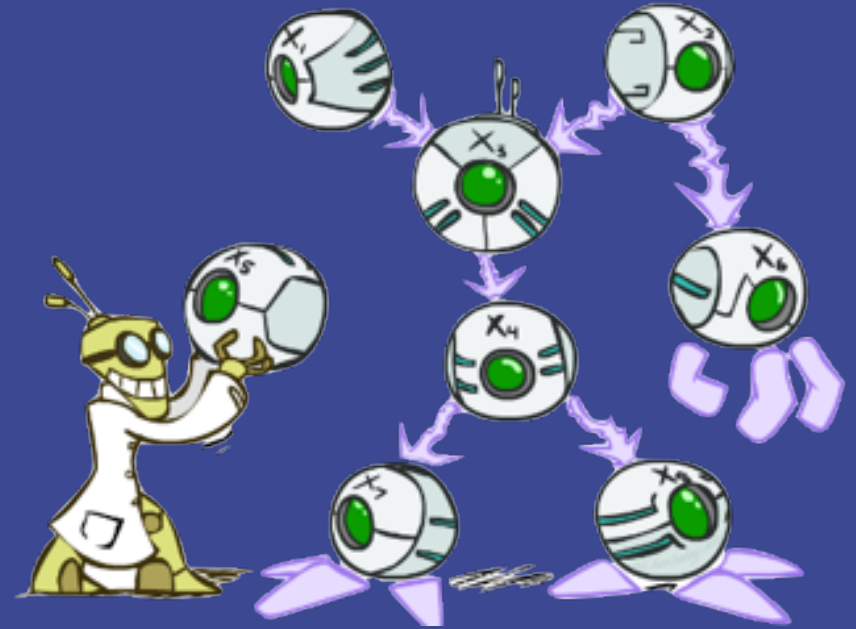


CS 156

Intro to AI

Bayesian Networks



These slides are based on the slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.
The artwork is by Ketrina Yim

Today: Bayes Nets

- ▶ Representation
- ▶ Independence in Bayes Nets
- ▶ Probabilistic Inference in Bayes Nets

Bayes Nets: Big Picture

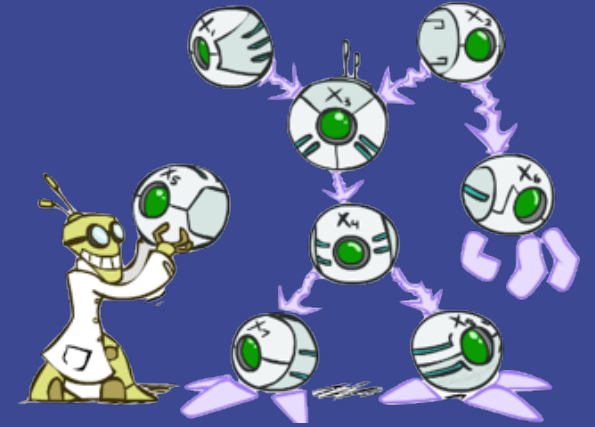
Two problems with using full joint distribution tables to represent our probabilistic models:

1. Unless there are only a few variables, the joint distribution is way too big to represent explicitly: for n variables with domain size d , the size of the joint distribution is d^n .
2. Hard to learn (estimate) joint probabilities empirically when more than a few variables are involved at a time.

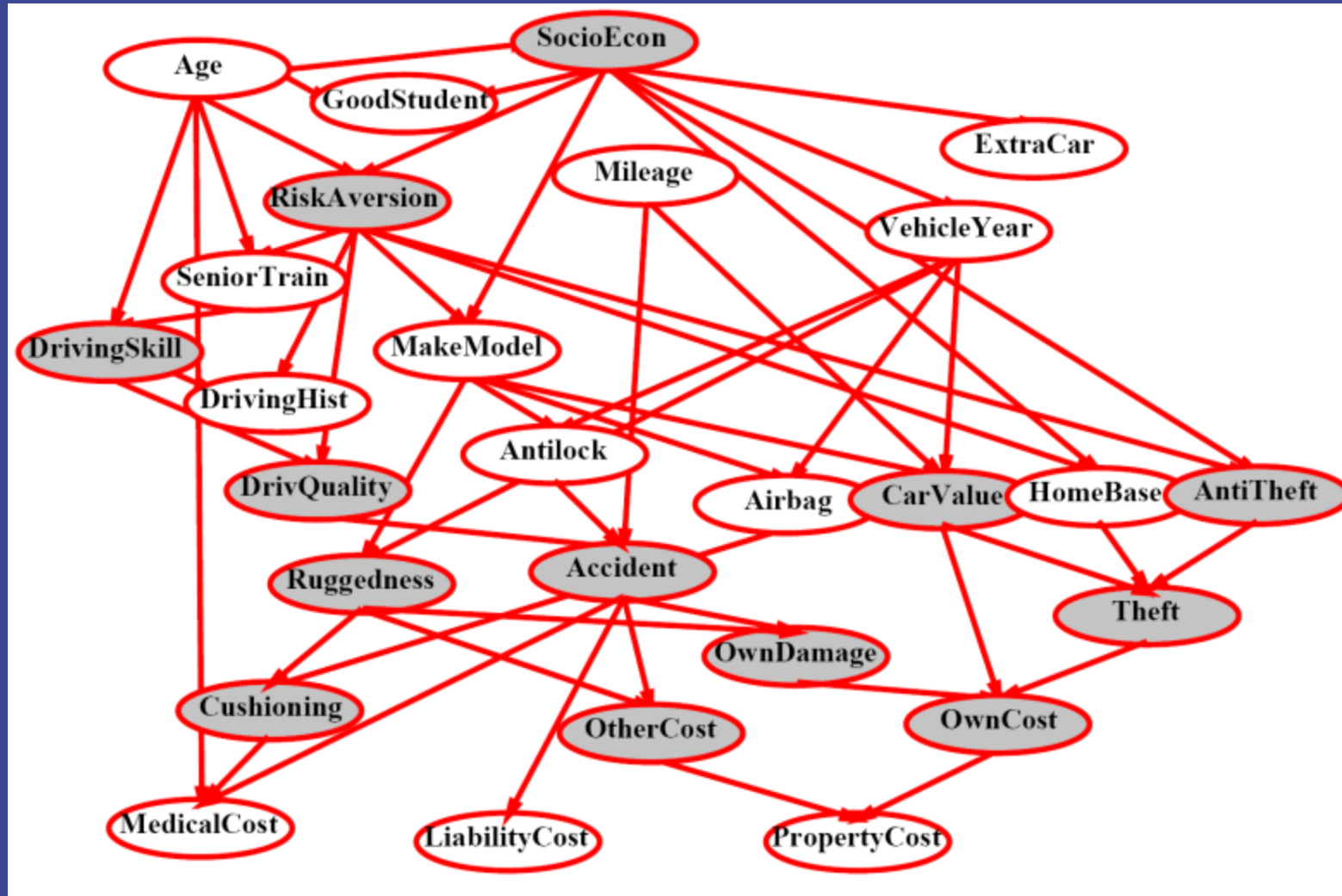
Bayes Nets: Big Picture

Bayes nets: a technique for describing complex joint distributions **compactly** using simple, local conditional distributions

- Subset of **graphical models**
- we describe how variables locally interact
- local interactions chain together to give global, indirect interactions

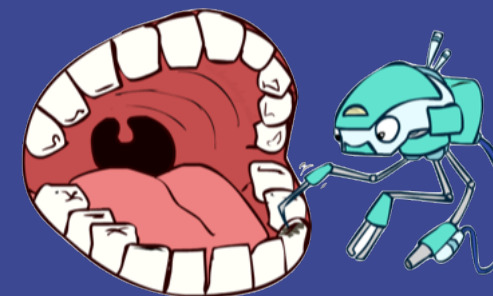
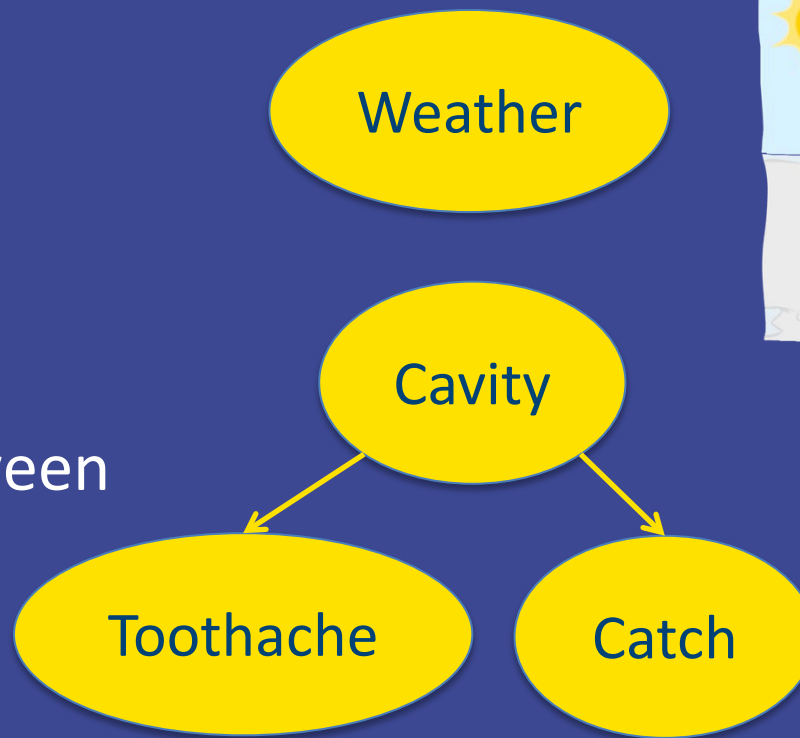


Example Bayes Net: Insurance



Graphical Model Notation

- ▶ Nodes: variables (with domains)
 - Can be assigned (observed) or unassigned (unobserved)
 - ▶ Arcs: interactions
 - Similar to CSP constraints
 - Indicate “**direct influence**” between variables
 - Formally: encode conditional independence (more later)
-
- For now: imagine that arrows mean direct causation



Example: Coin Flips



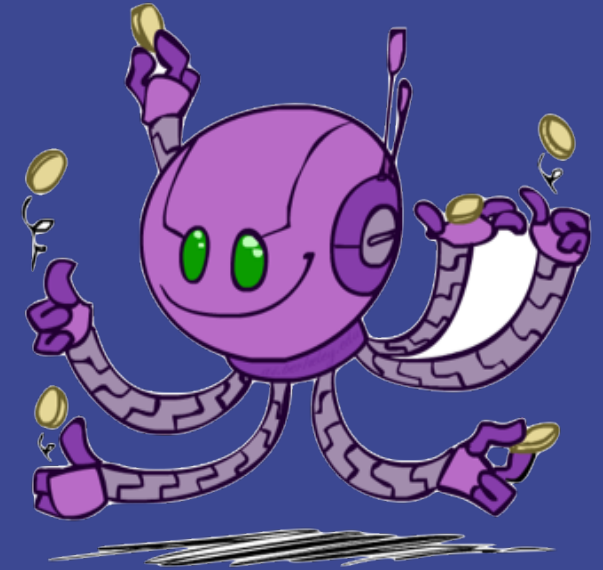
- ▶ N independent coin flips:
each coin flip represented by a random variable X_i

X_1

X_2

...

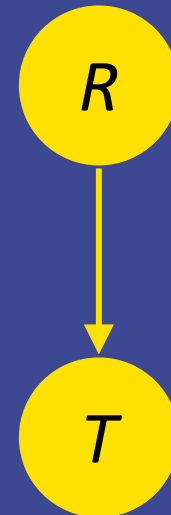
X_n



- ▶ No interactions between variables: **absolute independence**

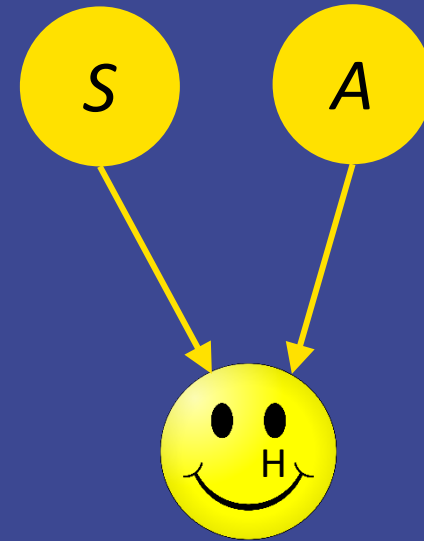
Example: Traffic

- ▶ Variables:
 - R: It rains
 - T: There is traffic
- ▶ Model: rain causes traffic



Example: Happiness

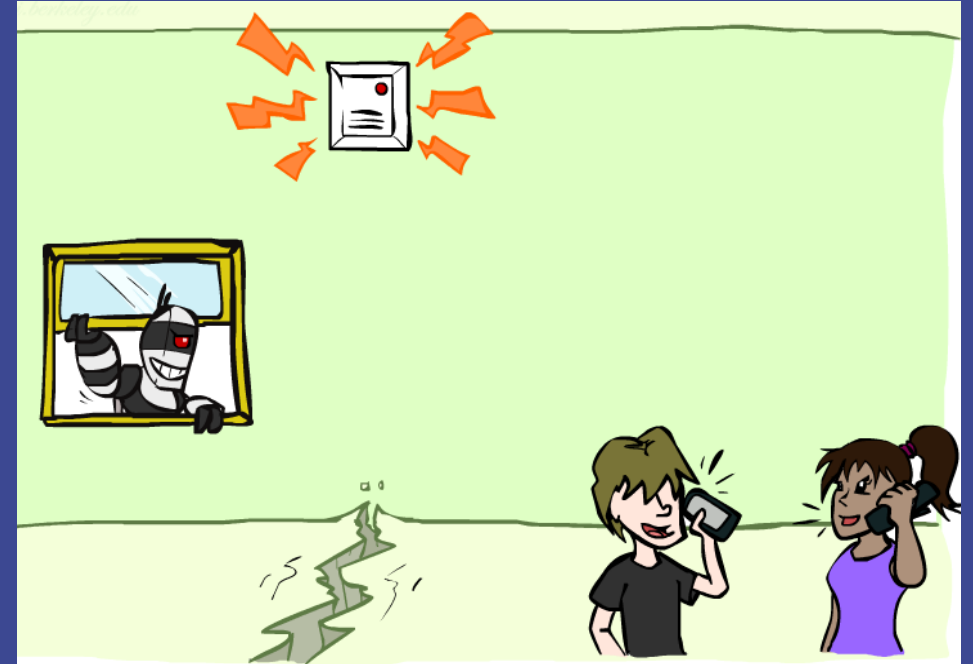
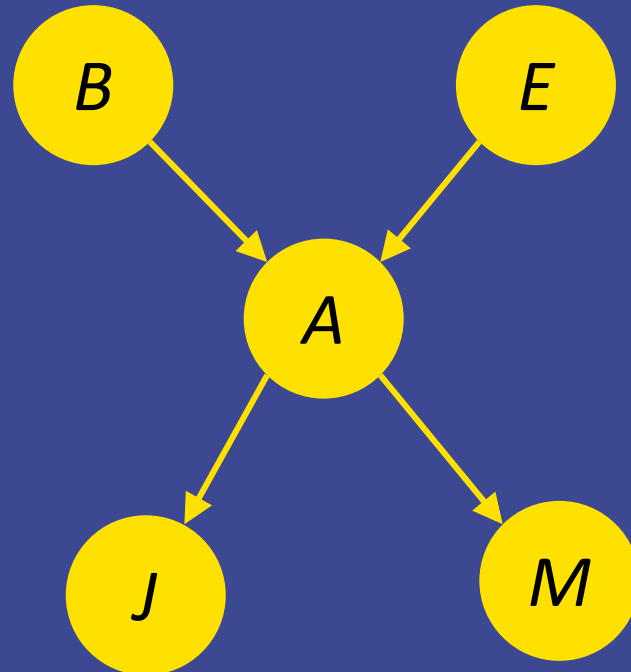
- ▶ Variables:
 - A: Anna gets an A
 - S: It is sunny
 - H: Anna is happy



Example: Alarm Network

► Variables

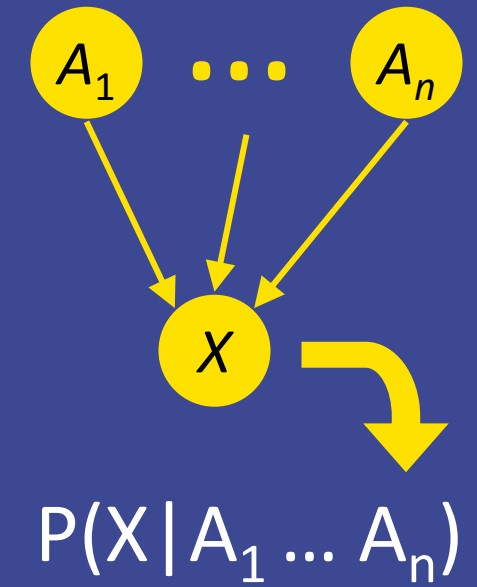
- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Bayes Net Semantics



- ▶ A set of nodes one node per random variable
 - ▶ Nodes are connected in a directed, acyclic graph
 - ▶ A conditional probability table (CPT) for each node
 - A collection of distributions over X , one for each combination of parents' values
- $$P(X|a_1 \dots a_n)$$



A Bayes net = Topology (graph) + Local Conditional Probabilities

Probabilities in BNs

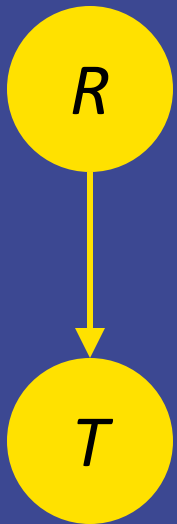


Bayes' nets **implicitly** encode joint distributions as a product of local conditional distributions.

To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$\begin{aligned} P(x_1, x_2, \dots, x_n) &= \prod_{i=1}^n P(x_i | \text{parents}(X_i)) \\ &= P(x_1 | \text{parents}(X_1)) P(x_2 | \text{parents}(X_2)) \dots P(x_n | \text{parents}(X_n)) \end{aligned}$$

Example: Traffic



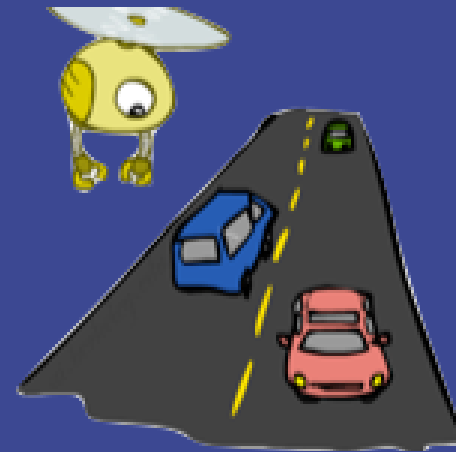
$P(R)$

+r	1/4
-r	3/4

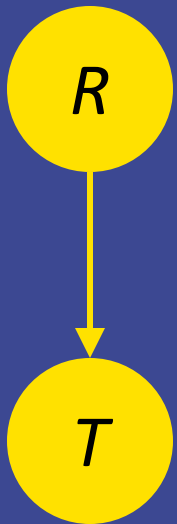
$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(+r, -t) =$$



Example: Traffic



$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

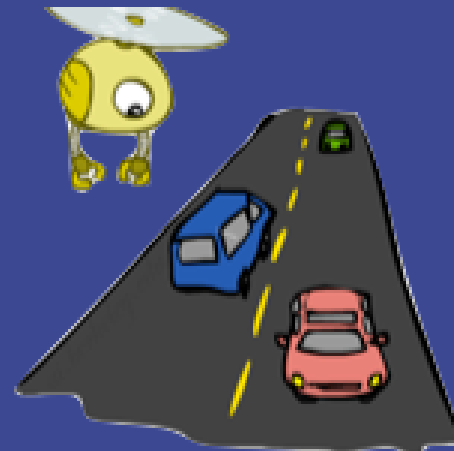
+r

+t	3/4
-t	1/4

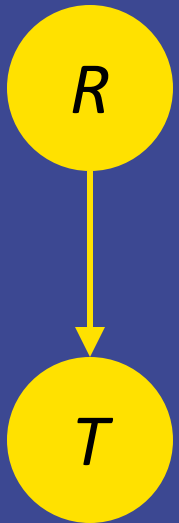
-r

+t	1/2
-t	1/2

$$P(+r, -t) = P(-t|+r) P(+r)$$



Example: Traffic



P(R)

+r	1/4
-r	3/4

P(T|R)

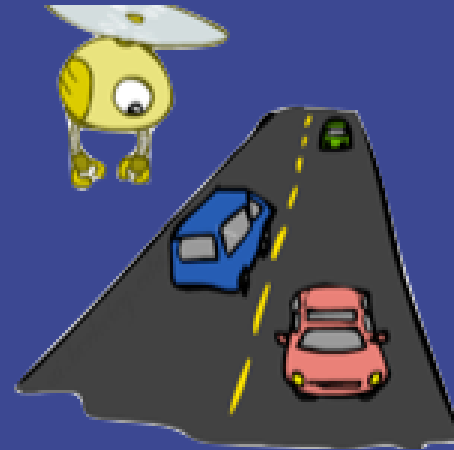
+r

+t	3/4
-t	1/4

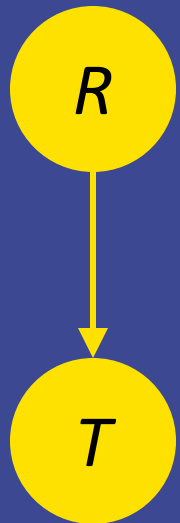
-r

+t	1/2
-t	1/2

$$P(+r, -t) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$



iClicker: Traffic



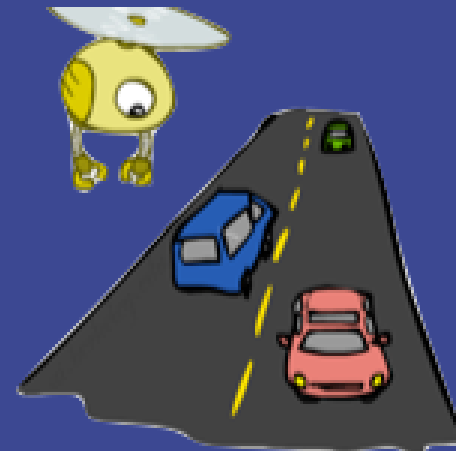
$P(R)$

+r	1/4
-r	3/4

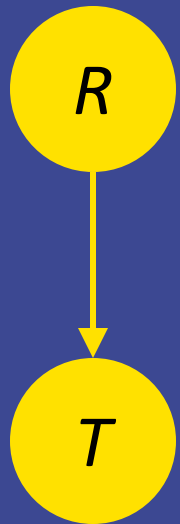
$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(-r, +t) =$$



iClicker: Traffic



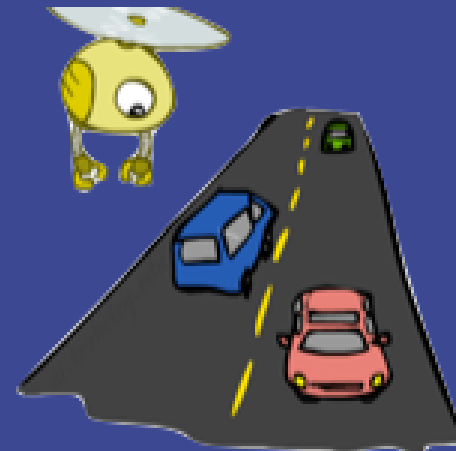
$P(R)$

+r	1/4
-r	3/4

$P(T|R)$

+r	+t	3/4
	-t	1/4
-r	+t	1/2
	-t	1/2

$$P(-r, +t) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$$



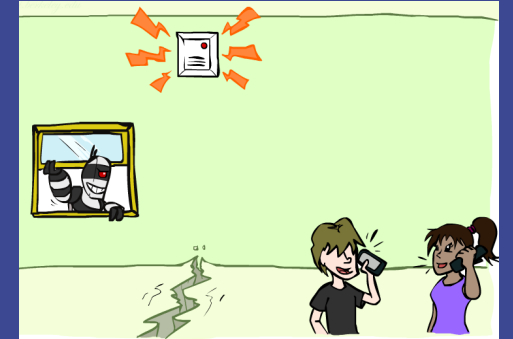
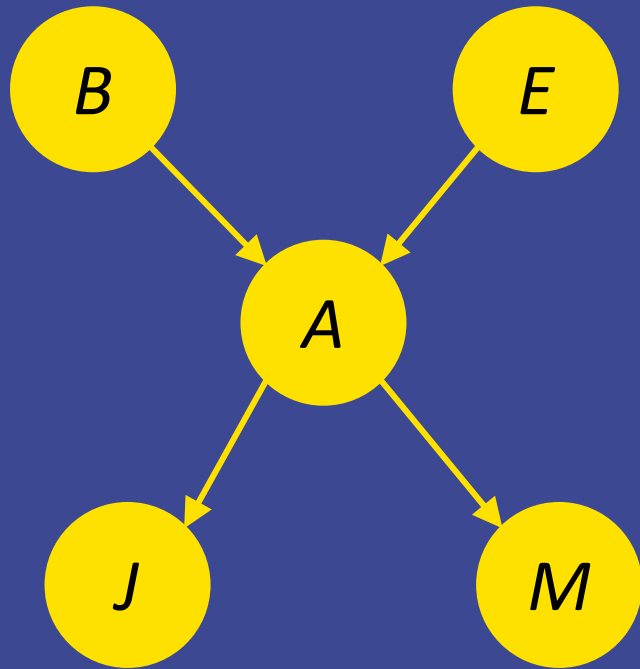
Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

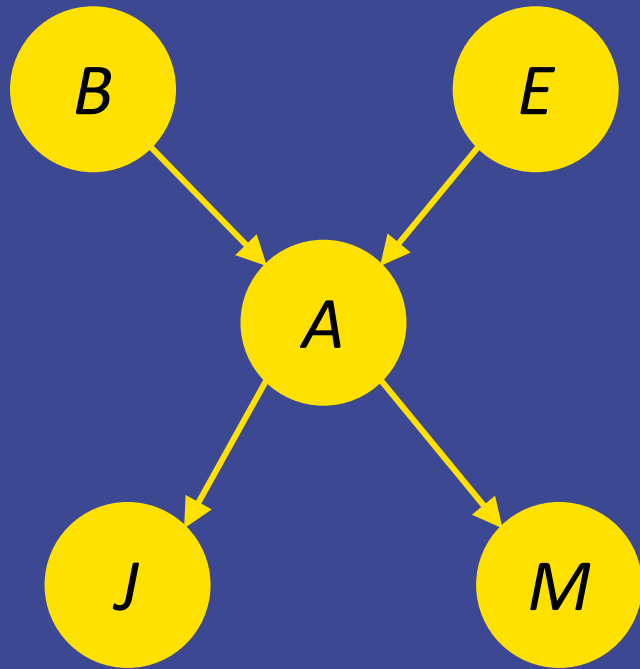


$P(+b, -e, +a, -j, +m) =$

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Example: Alarm Network

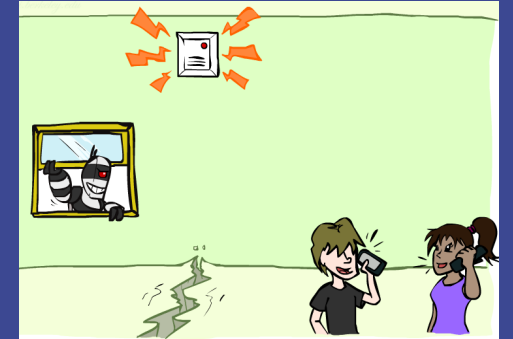
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-b	0.999



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+a	-m	0.3
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-a	-m	0.99



B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 &P(+b, -e, +a, -j, +m) = \\
 &P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) = \\
 &0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7
 \end{aligned}$$

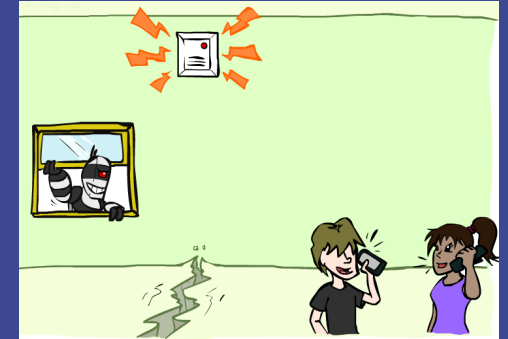
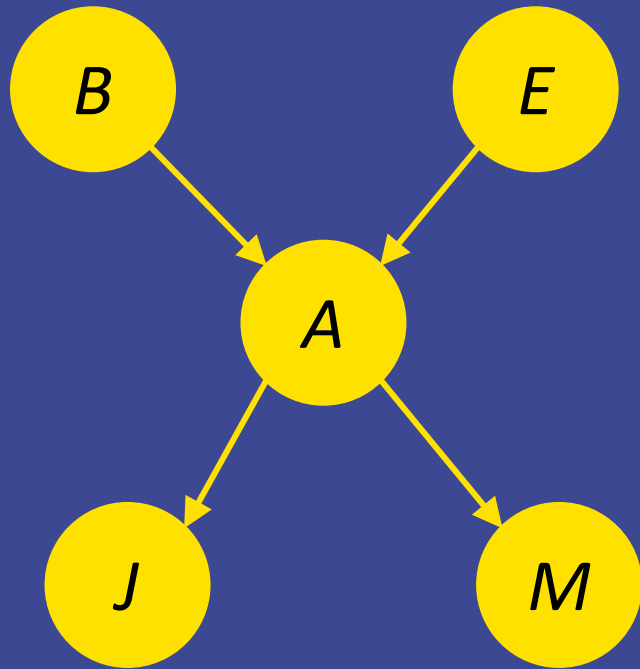
iClique: Alarm Network

B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
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-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99



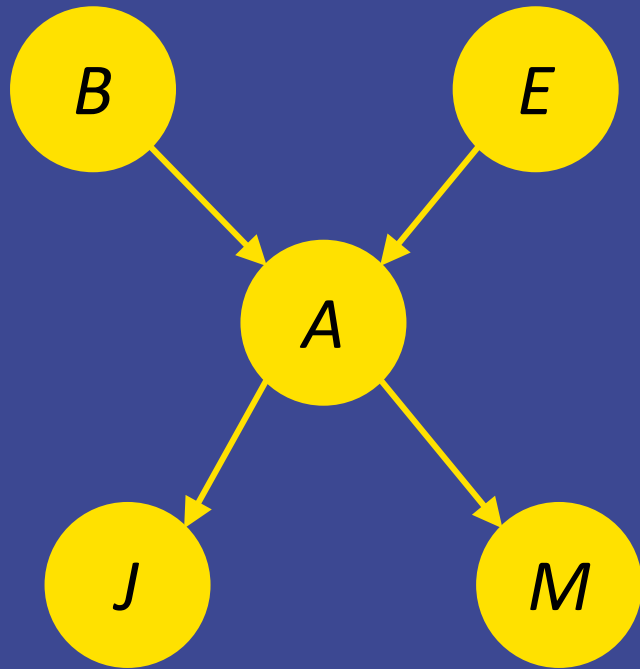
B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$P(+b, +e, +a, -j, -m) = ?$

Answer format: ddx10⁽⁻ⁿ⁾

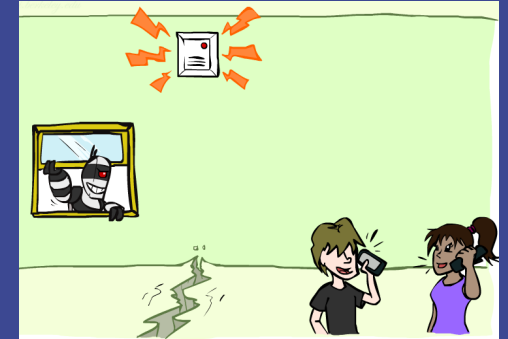
iClique: Alarm Network

B	P(B)
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E	P(E)
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-e	0.998

A	M	P(M A)
+a	+m	0.7
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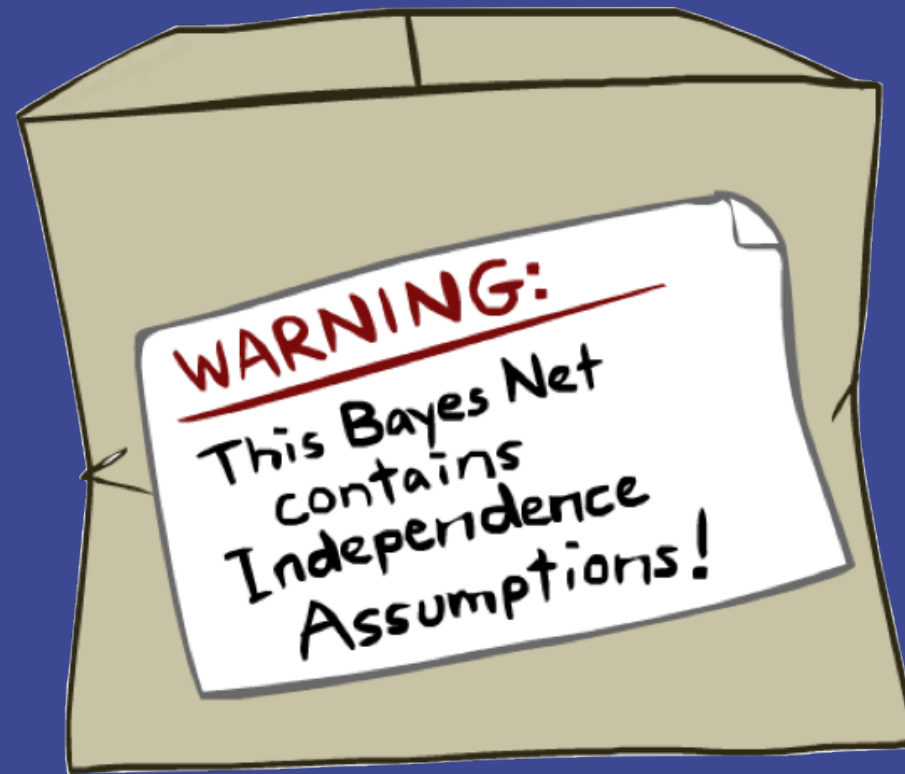
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+b	+e	+a	0.95
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+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

$$\begin{aligned}
 &P(+b, +e, +a, -j, -m) = \\
 &P(+b)P(+e)P(+a|+b, +e)P(-j|+a)P(-m|+a)= \\
 &0.001 \times 0.002 \times 0.95 \times 0.1 \times 0.3 = 57 \times 10^{-9}
 \end{aligned}$$

Size of a Bayes' Net

- ▶ How big is a joint distribution over N Boolean variables?
 2^N
- ▶ How big is an N -node net if nodes have up to k parents?
 $O(N * 2^{k+1})$
- ▶ Both give us the power to calculate $P(X_1, X_2, \dots, X_n)$
- ▶ Bayes Nets: Huge space savings!
- ▶ Easier to elicit local CPTs
- ▶ Faster to answer queries

Bayes Nets: Assumptions



Causal Chains

This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x) P(y|x) P(z|y)$$

Z independent of X ?

No!

Causal Chains

This configuration is a "causal chain"



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x) P(y|x) P(z|y)$$

Z independent of X given Y?

$$P(z|x, y) =? P(z|y)$$

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)}$$

$$= P(z|y)$$

Yes!

Evidence along the chain “blocks” the influence

Common Cause

This configuration is a "common cause"

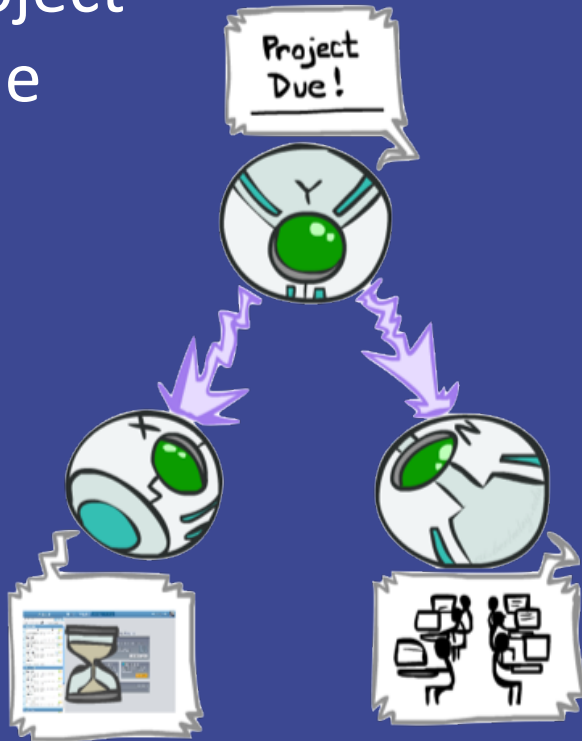
Z independent of X ?

No!

Y: Project
due

X:
Forum
busy

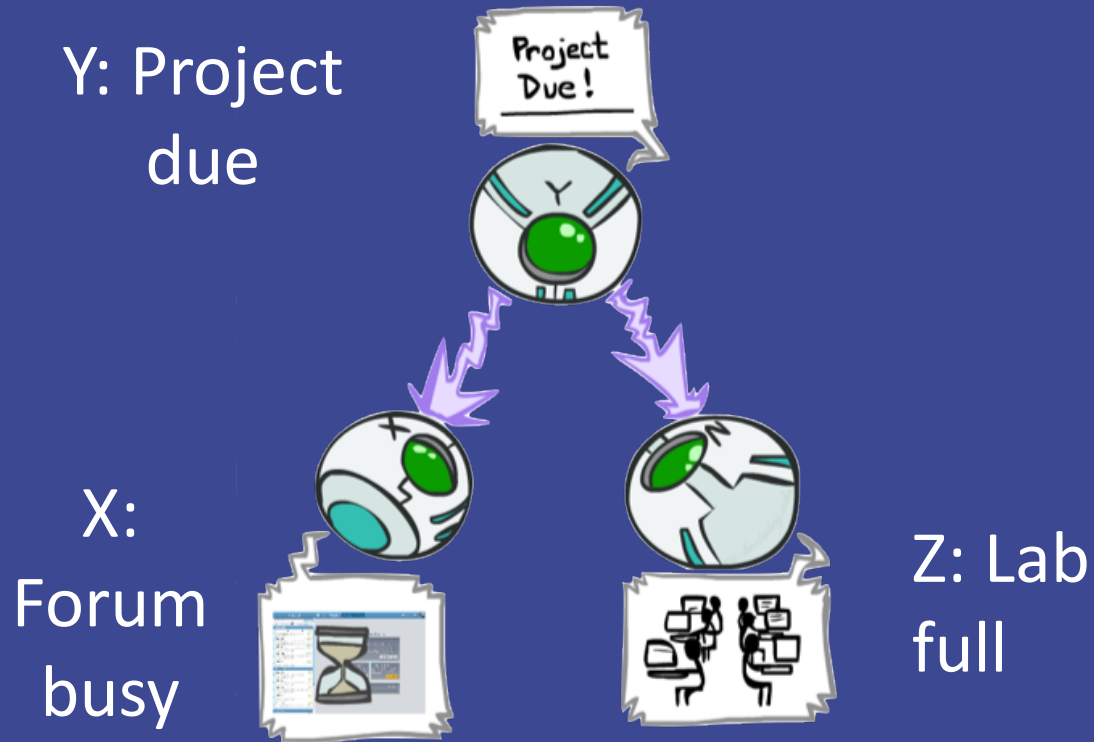
Z: Lab
full



$$P(x, y, z) = P(y) P(x|y) P(z|y)$$

Common Cause

This configuration is a "common cause"



$$P(x, y, z) = P(y) P(x|y) P(z|y)$$

Z and X independent given Y?

$$P(z|x, y) =? P(z|y)$$

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)}$$

$$= P(z|y)$$

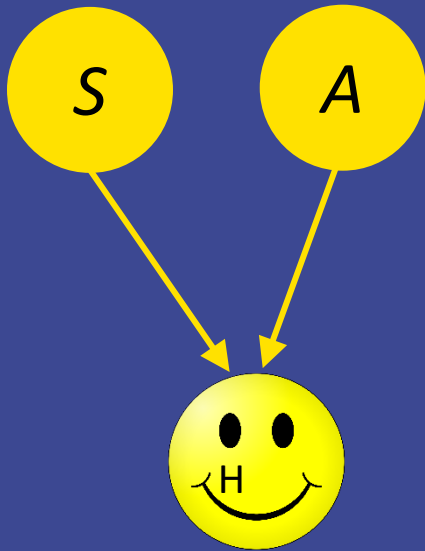
Yes!

Observing the cause blocks influence between effects

Common Effect

Common effect (v-structure)

X: It is **S**unny Y: Anna gets **A**



Z: **H**appiness

Are X and Y independent?

Yes: they both make Anna happy, but they are not otherwise related

Are X and Y independent given Z?

No: Anna's happiness puts the grade and the weather in causal competition.

Observing an effect activates influence between possible causes.

The General Case



The General Case

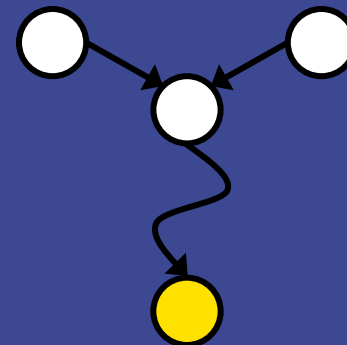
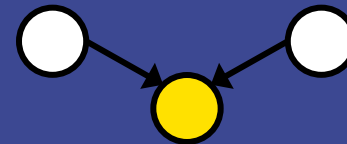
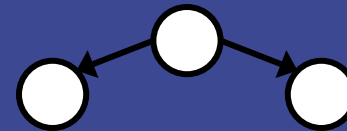
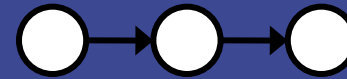
- ▶ General question: in a given BN, are two variables independent (given evidence)?
- ▶ Solution: break down the graph into segments
- ▶ Any complex example can be broken into repetitions of the three canonical cases

Active / Inactive Paths

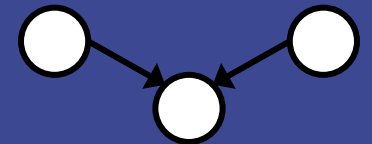
A triple is active:

- ▶ Causal chain $X \rightarrow Y \rightarrow Z$ where Y is unobserved (either direction)
- ▶ Common cause $X \leftarrow Y \rightarrow Z$ where Y is unobserved
- ▶ Common effect (aka v-structure)
 $X \rightarrow Y \leftarrow Z$ where Y or one of its descendants is observed

Active Triples



Inactive Triples

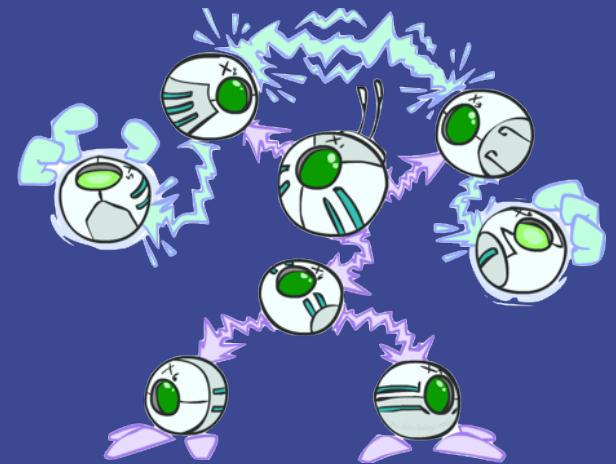


Active / Inactive Paths

- ▶ Are X and Y conditionally independent given evidence variables Z?
 - Consider **all** (undirected) paths from X to Y
 - **No active paths** = independence! We say that X and Y are "d-separated" by Z
 - If one or more active path? We cannot guarantee independence
- ▶ A path is active if each segment/triple is active
- ▶ **All it takes to block a path is a single inactive segment (triple)**

General Case

- Query: $X_i \perp\!\!\!\perp X_j \mid \{X_{k1}, \dots X_{kn}\}$?
- Check all **undirected** paths between X_i and X_j
 - If one or more active, then independence **is not guaranteed**:
 - Otherwise (**if all paths are inactive**), independence is guaranteed



Independence

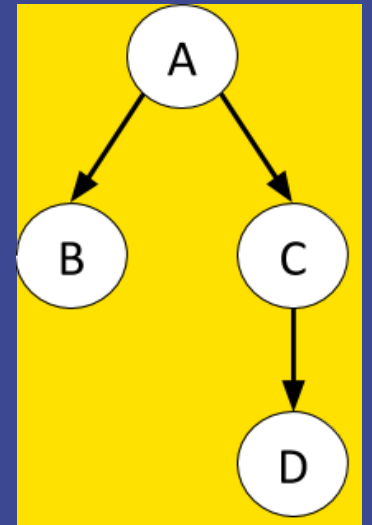
$A \perp\!\!\!\perp D \mid B$?

Consider all (undirected) paths from A to D:

- ▶ only one: ACD

The path includes one triple only (ACD)

- ▶ ACD is active 
- ▶ Active path => No independence



Independence

$B \perp\!\!\!\perp D \mid A ?$

Consider all (undirected) paths from B to D:

- ▶ only one: BACD

The path includes triples BAC and ACD

- ▶ BAC is inactive

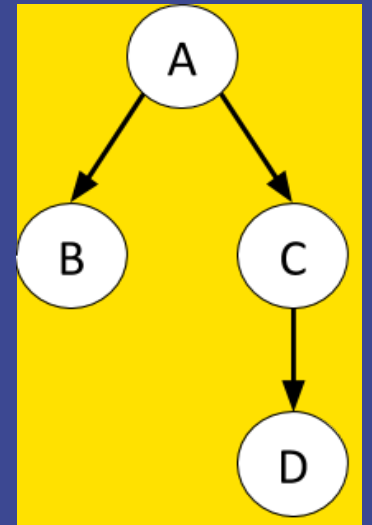


All it takes to block a path is a single inactive segment

=> BACD is inactive

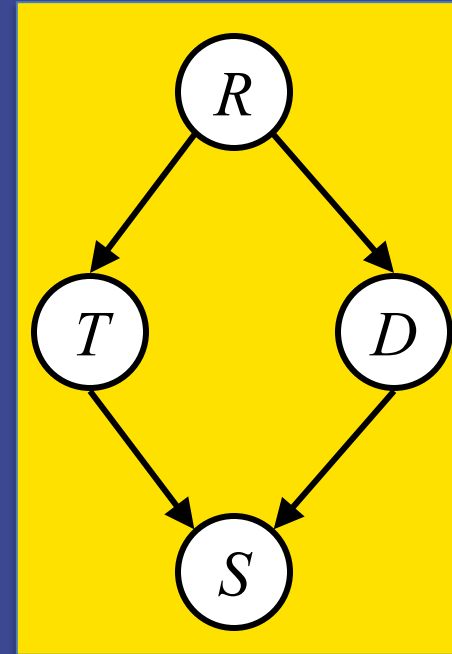
No active paths => independence guaranteed

$B \perp\!\!\!\perp D \mid A$



Independence

- ▶ Variables:
 - R: Raining
 - T: Traffic
 - D: Roof drips
 - S: I'm sad

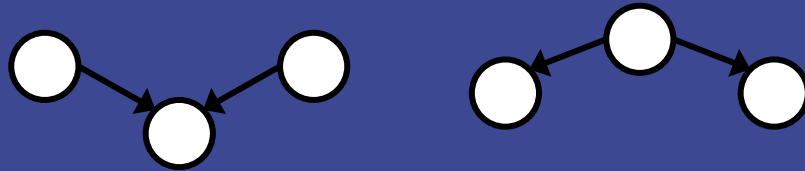


Independence

$T \perp\!\!\!\perp D$?

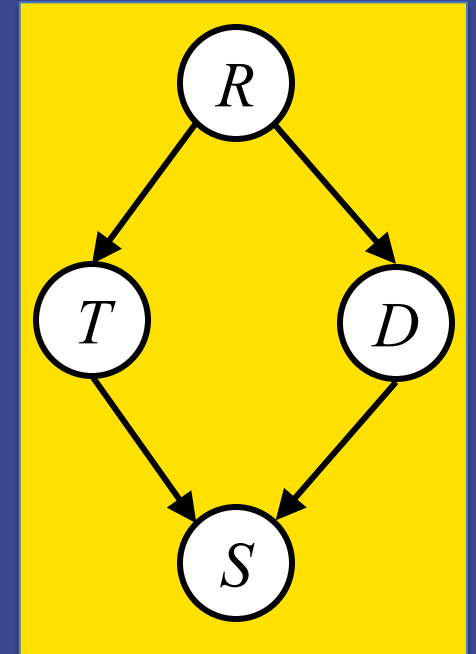
Consider all (undirected) paths from T to D

There are two paths: TSD and TRD



TSD inactive

TRD active => *No independence*

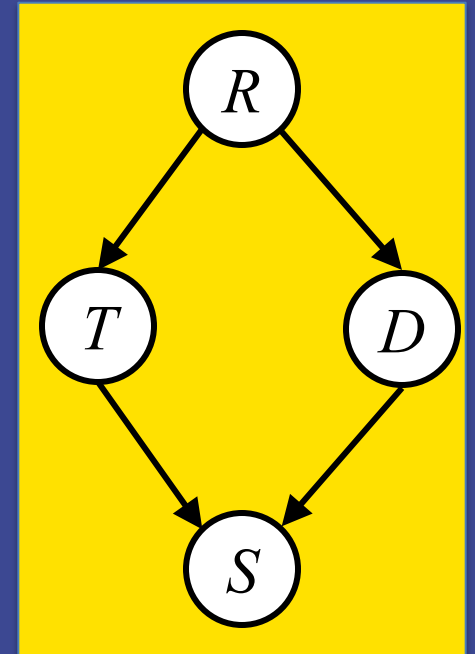


iClicker: Independence

$T \perp\!\!\!\perp D \mid R$?

iClicker:

- A. Yes, we can guarantee independence
- B. No, we cannot guarantee independence

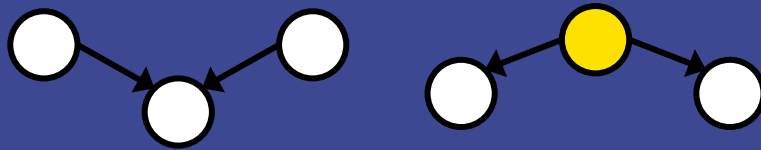


Independence

$T \perp\!\!\!\perp D \mid R?$

Consider all (undirected) paths from T to D

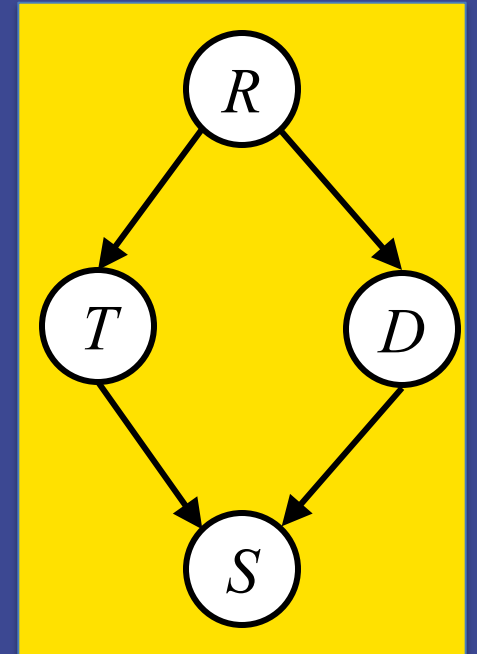
There are two paths: TSD and TRD



TSD inactive

TRD inactive

No active paths $\Rightarrow$ independence



Independence

$T \perp\!\!\!\perp D \mid R, S?$

Consider all (undirected) paths from T to D

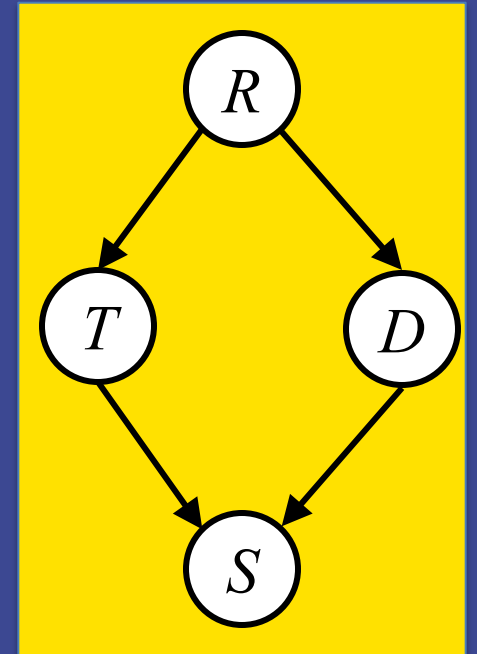
There are two paths: TSD and TRD



TSD active $\Rightarrow$ *No independence*

TRD inactive

One active path $\Rightarrow$ No independence



Probabilistic Inference

- ▶ General case:
 - Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query variable(s) : Q
 - Hidden variables: $H_1 \dots H_k$
 - We want: $P(Q | E_1 = e_1, \dots, E_k = e_k)$

Probabilistic Inference in Bayes' Net

▶ John and Mary called.

▶ Was there a burglary?

▶ Query Variable

- B

▶ Evidence Variable

- J, M

▶ Hidden Variable

- E, A

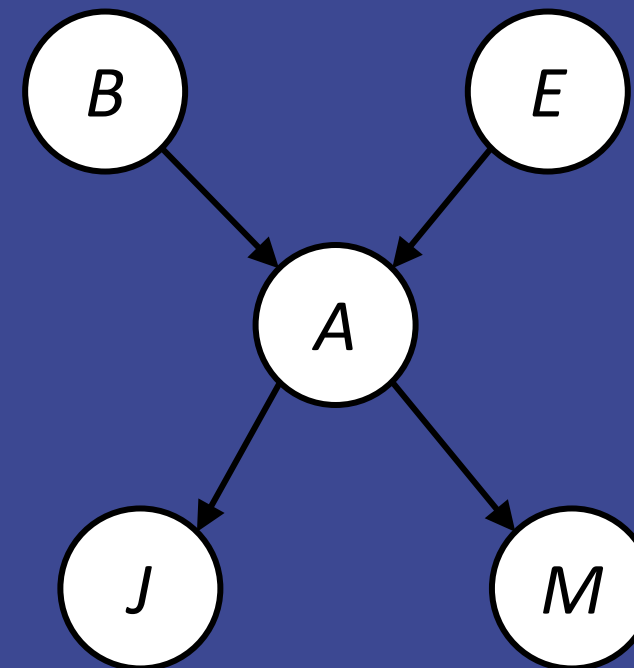
What probability are we looking for?

A. $P(+b, +j, +m)$

B. $P(+b \mid +j, +m)$

C. $P(+b, +j, +m, -e, +a)$

D. $P(+b, +j, +m \mid -e, +a)$



Next

- ▶ Question of the Week on Tuesday
- ▶ Inference by enumeration in Bayes Nets
- ▶ Probabilistic Reasoning over time
 - Markov Chains
 - Hidden Markov Models